

Aadhishrii Patiil

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Education

University of Southern California, Los Angeles, USA (MS. in Computer Science) Jan. 2023 – Dec. 2024

Related Courses: Algorithms, Database Management Systems, Web Technologies, Machine Learning, Information Retrieval

University of New South Wales, Sydney, Australia (BE. in Software Engineering) Aug. 2018 – Dec. 2022

Related Courses: Data Structures, Object Oriented Programming, Software Design, Operating Systems, Artificial Intelligence

Work Experience

University of Southern California Feb. 2025 – Present

Research Associate - Software Engineer | Next.js, GraphQL, Cloudflare, Supabase, Llama-3

- Architected Terraform-managed serverless analytics platform **ingesting 50k** papers monthly, detecting emerging topics within 15 minutes
- Built **Cloudflare Workers / Supabase Functions** microservices; ETL with backoff & **GitHub Actions** ensured 99.9% overall uptime
- Implemented **pgvector semantic search**; tuned **IVFFlatANN**, reducing retrieval misses by 92% and yielding **sub-50ms queries**
- Integrated **Llama-3 GPU summarization** via Workers AI; produced concise 150-word briefs, halving researchers' literature-survey time

Trini AI Jun. 2024 – Sep. 2024

Software Engineer - Intern | JavaScript, React, FastAPI, PostgreSQL, Git

- Architected and deployed an AI-driven legal document automation platform for law firms, enabling **200+ contracts/day** to be classified and parsed using React, FastAPI, PostgreSQL, and LegalBERT for automatic key term extraction and clause matching
- **Reduced review time by 25%** by developing accessible, **modular React components**—including upload flow, summary cards, and review modals—aligned with **WCAG 2.1 standards** and tested through simulating real-world case documents
- Validated component reliability through **50+ unit and integration tests** using **React Testing Library** and **Jest**, achieving **90%+ code coverage** by testing dynamic rendering, accessibility behaviors, and upload/review edge cases
- Facilitated agile delivery across **6 sprints** by merging **40+ PRs**; coordinating closely with backend engineers & product designers

Royal Hospital for Women (UNSW) Sep. 2021 – Aug. 2022

Software Engineer | JavaScript, TypeScript, React, Figma, Git

- Engineered a **modular React frontend** for NICU discharge coordination using **controlled components, custom form hooks, schema-based validation, & conditional rendering** to manage user states and ensure robust data capture across clinical workflows
- Led iterative usability testing with **Figma prototypes** and **task-based evaluation**, refining navigation, data entry, and alert flows
- Defined requirements through **stakeholder interviews** and **user journeys**, delivering **accessible UIs** aligned with usability standards
- Improved transfer workflows by **15%** via **component-based UI** features for vitals tracking, caregiver training, and emergency escalation

Technical Skills

Programming Languages Python, Java, JavaScript, TypeScript, C/C++, HTML5/CSS3, PHP

Web/Database Technologies ReactJS, Angular, NodeJS, Spring-Boot, Bootstrap, PostgreSQL, MySQL, Flask, jQuery, MongoDB, GraphQL

Libraries/Frameworks NumPy, Pandas, Scikit-learn, Seaborn, TensorFlow, Large Language Models (Llama-3), Linux, VIM

Tools & Platforms Amazon Web Services (AWS), GCP, Cloudflare, Supabase, Kafka, Figma, Git, Swagger, Postman, Docker

Functional Skills Data Science, Machine Learning, CI/CD, Agile-Scrum, Microservice Architecture, Web Development

Technical Projects

TrackFlow: Distributed Order/Event Pipeline | Spring Boot, Apache Kafka, PostgreSQL (AWS RDS), AWS S3, AWS EC2

- Developed a distributed order pipeline using **Spring Boot** and **Kafka** to simulate event-driven order lifecycle stages across services
- Persisted order state transitions in **AWS RDS (PostgreSQL)**, ensuring real-time consistency and durability of service communication
- Integrated **AWS S3** for uploading and managing order invoices programmatically through **Spring Boot** and **AWS SDK**
- Deployed Spring Boot microservices on **AWS EC2** using networking and setup scripts to enable service communication and testing

MarketPulse: Real-Time Stock Market Analytics | React.js, Angular.js, Node.js, Google Cloud Platform (GCP)

- Built interactive stock analytics dashboard using **Angular**, integrating **FinnHub & Polygon.io** for real-time financial data visualization
- Deployed on **GCP** with **auto-scaling & load balancing**, ensuring smooth performance during simulated trading scenarios & live updates
- Designed **RESTful Node.js backend** to aggregate API data, improving reliability & reducing third-party call volume through **caching**
- Streamlined **cloud deployment** by containerizing services with **Docker** and automated it using **GCP Cloud Run & Cloud Build** tools

Breeze: Smart Event Planning with Safety Insights | Next.js, Flask, PostgreSQL, Apache Kafka

- Developed responsive **Next.js frontend** with dynamic routing and state-managed components for event setup and invitation links
- Built **RESTful Flask backend** for events, users, and responses; **optimized PostgreSQL** with **indexing** and **connection pooling**
- Streamed COVID updates via **Kafka**, triggering real-time notifications when Aus Gov API flagged elevated location-based risk levels
- Implemented **form-based** fallback for undecided venues using **questionnaire scoring** and external API to suggest low-risk locations